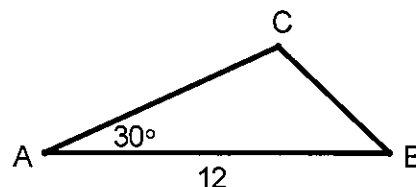


8. The angle of elevation of the summit from the base of a ski lift is 35° . If a skier rides 1000 feet on this lift to the summit, find the number of feet in the **vertical distance** between the base of the lift and the summit. Express your answer as a **decimal** rounded to the nearest tenth of a foot.

9. In the diagram, $\angle CAB = 30^\circ$, and $AB = 12$. $AC = k\sqrt{3}$, and $BC = w$ where k and w are positive integers, and $w < 164$. Find the sum of all possible distinct values of w .



10. If $A = \begin{bmatrix} 2 & 3 \\ 1 & -3 \end{bmatrix} \bullet \begin{bmatrix} 5 & 4 \\ 1 & 7 \end{bmatrix}$, find the value of $\det(A)$.

11. If $(3x - 4y + z + 2a)^2$ is expanded and completely simplified, find the sum of the numerical coefficients.

12. Let x be an integer such that $0 < x < 150$. Find the sum of all possible distinct values of x such that all three of the following conditions hold: $\tan(3x + 17)^\circ < 0$, $\sin(5x - 13)^\circ > 0$, and $\cos(2x + 3)^\circ < 0$.

13. The area of the ellipse $\frac{(x-3)^2}{36} + \frac{(y+2)^2}{k} = 1$ is 18 times the area of the circle $x^2 + y^2 = 4$.

Find the value of k .

14. In the diagram, $\triangle DEF$ is equilateral, points A , B , and C lie on $\overline{EF}$, $\overline{DF}$, and $\overline{DE}$ respectively. $\overline{EA} \cong \overline{FB} \cong \overline{DC}$. The ratio of the area of $\triangle ABC$ to the area of $\triangle DEF$ is $53:128$. If $EF = 256$, and $AE < AF$, find AF . Express your answer as a decimal rounded to the nearest tenth.

